

Question number	Answer	Additional guidance	Mark
5(a)(i)	A description that includes three of the following points: • fibrous protein (1) • (protein) composed of {three polypeptide chains / three-stranded / triple} <u>helix</u> (1) • held by hydrogen bonds (between the chains) (1) • credit details of the chains (1)	e.g every third amino acid is a glycine, repeating sequences of amino acids, high content of {glycine / proline / hydroxyproline}	(3)

Question number	Answer	Additional guidance	Mark
5(a)(ii)	An explanation that includes the following points: • gives (the wall) (tensile) strength (1) • so that the aorta {does not get damaged by / can withstand} pressure (of the blood leaving the heart) (1)	IGNORE refs to elastic properties and recoil IGNORE prevents aorta from collapsing	(2)

Question number	Answer	Additional guidance	Mark
5(b)	Indicative content: Graph 1 - older male monkeys have more stiffness than younger males (D) - older female monkeys have more stiffness than younger females (D) - older males have more stiffness than older females (D) - probably significant as error bars do not overlap (C) - not much difference in stiffness between younger males and females (D) - as error bars overlap (C) - age increases aortic stiffness in both males and females (C) - age has a greater effect on aortic stiffness in males than females (C) Graph 2 - density of collagen decreases slightly with age (D) - males of all ages have (slightly) more collagen than females (C) - probably not significant as error bars overlap (C) - neither age nor sex affects density of collagen (C) - changes in stiffness do not appear to be related to the density of collagen (C) Graph 3 - higher type 1 in younger monkeys than older ones (D) - more type 1 in females than males at each age (D) - no error bars shown to judge significance (C) Graph 4 - more type 8 in older male monkeys (D) - may not be a difference in type 8 between older and younger females (D) - no error bars to judge significance (C) - the type of collagen appears to determine stiffness (C) - stiffness associated with decrease in type 1 and increase in type 8 (C)	Level 1 1 mark = description made from one graph (D) 2 marks = descriptions made from two graphs (D) Level 2 3 marks = plus one conclusion made (C) 4 marks = plus two conclusions made (C) Level 3 5 marks = two conclusions and comments on the other two graphs 6 marks = three conclusions that includes the asterisked conclusion (C*) and comments on the other two graphs Description = comparison of one variable Conclusion = summary statement that includes both age and sex interpretation of error bar and significance of data – not reliability links between 2 graphs	(6)